

Math GRE Prep Day 2

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Outline

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Problem 1

Suppose that $f(x)$ is twice differentiable with $f(2) = 10$ and $f'(x) = x^2 f(x)$. Find $f''(2)$.

Problem 2

Find

$$\lim_{x \rightarrow 0} \frac{\sin(3x) \sin(2x)}{x \tan(6x)}.$$

Problem 3

How many tangent lines to the curve $y = x^2$ pass through the point $(3, -7)$ and what are their slopes?

Problem 4

The tangent line to the curve $y = f(x)$ at $x = 2$ is $y - 8 = 2(x - 2)$. Calculate

$$\lim_{x \rightarrow 0} \frac{xf(2) - 2f(x)}{x - 2}$$

Problem 5

Suppose that $f(x) = |x - 1| - |x + 1|$. Compute $\int_{-3}^3 (f(x - 1) - f(x + 1) + 1) dx$

Problem 6

Find an integral expression which is equal to

$$\lim_{n \rightarrow \infty} \frac{3}{n^2} \left(\sqrt{4n^2 + 9 \times 1^2} + \sqrt{4n^2 + 9 \times 2^2} + \cdots + \sqrt{4n^2 + 9 \times (n-1)^2} \right)$$

Problem 7

A region is bounded by two concentric circles, where the smaller circle has radius r and the bigger circle has radius R . If r decreases at 1 inch per second and R increases at 2 inches per second, how fast is the area changing at the instant when $r = 3$ inches and $R = 4$ inches?

Problem 8

Let $f(x) = x^{2024}$. Evaluate

$$f(1) + \frac{f'(1)}{1!} + \frac{f''(1)}{2!} + \cdots + \frac{f^{(2024)}(1)}{2024!}$$

Problem 9

Find the smallest positive integer m such that $m^{1/m} \geq n^{1/n}$ for all positive integers $n \geq m$.

Problem 10

Find all values of x such that

$$\lim_{n \rightarrow \infty} (x + x^3 + x^5 + \cdots + x^{2n-1}) = \frac{3}{8}.$$

Outline

Problem 11

Evaluate

$$\lim_{n \rightarrow \infty} \left(\sqrt{1 + 2 + 3 + \cdots + n} - \sqrt{1 + 2 + 3 + \cdots + (n-1)} \right)$$

Problem 12

Evaluate

$$\lim_{n \rightarrow \infty} \frac{\sqrt{1 \cdot 2} + \sqrt{2 \cdot 3} + \cdots + \sqrt{n(n+1)}}{n^2}$$

Problem 13

Evaluate

$$\sum_{n=1}^{\infty} \frac{1}{2^n} \sin \frac{n\pi}{3}.$$

Problem 14

Evaluate

$$\sum_{n=1}^{\infty} \frac{1}{2^n} \sin \frac{n\pi}{3}.$$

Problem 15

If $f(x) = e^{x^3} + e^{-x^3}$, evaluate $f^{(2022)}(0)$

Problem 16

Calculate

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{\sqrt{1^3 + 2^3 + \dots + i^3}}$$

Problem 17

Calculate

$$\lim_{x \rightarrow \infty} (\sin \sqrt{x + \pi} - \sin \sqrt{x})$$

Problem 18

Calculate

$$\lim_{n \rightarrow \infty} \cos \frac{1}{2} \cos \frac{1}{4} \cos \frac{1}{8} \dots \cos \frac{1}{2^n}$$